

# UNIVERSITY OF ENGINEERING AND TECHNOLOGY PESHAWAR

Department of Electrical Engineering

## LABORATORY REPORT

|                   |                                  |
|-------------------|----------------------------------|
| <b>Subject:</b>   | EE-170L Computer Programming Lab |
| <b>Lab No:</b>    | Lab 13                           |
| <b>Lab Title:</b> | Structures in C++                |
| <b>Semester:</b>  | Semester 2                       |
| <b>Session:</b>   | Spring 2026                      |

**Submitted By:**

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## 1. OBJECTIVE

To learn and apply structures in C++ including declaration, initialization, member access using dot operator, and copying structure variables. To understand how structures group related data of different types under a single name.

## 2. THEORY

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A structure is a user-defined data type that groups related variables of different data types under a single name. Unlike arrays where all elements must be the same type, structures can contain members of various types (int, float, char, string, etc.).

Key Concepts: (1) Structure Declaration: `struct StructureName { dataType member1; dataType member2; };` (2) Structure Variable Declaration: `StructureName variableName;` (3) Accessing Members: Using dot operator (`.`): `variableName.memberName` (4) Structure Initialization: `StructureName var = {value1, value2, ...};` (5) Structure Assignment: One structure variable can be assigned to another: `var2 = var1;`

Important features: The closing brace must be followed by a semicolon. Structures can be nested within other structures. Structure variables can be passed to functions. A structure declaration defines the 'form' but does not reserve memory until a variable is declared.

## 3. LAB TASKS

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### 3.1 Structure Declaration and Initialization (Person)

Declare a structure called Person with the following members: name (string), age (integer), address (string). Declare a variable called person1 of type Person and initialize its members with values of your choice. Display the values of the members of person1 using dot operator.

#### Source Code:

```
#include <iostream>
#include <string>
using namespace std;

// =====
// Structure Declaration
// =====
// Person structure to store personal information
// struct keyword is used to declare a structure
struct Person
{
    string name;        // Member to store person's name
    int age;            // Member to store person's age
    string address;     // Member to store person's address
}; // Semicolon is REQUIRED after structure declaration

int main()
{
```

```

// =====
// Structure Variable Declaration and Initialization
// =====
Person person1;    // Declare a variable of type Person

// Initialize members using dot operator (.)
person1.name = "Muhammad Adil";    // Assign name
person1.age = 20;    // Assign age
person1.address = "UET Peshawar, Pakistan"; // Assign address

// =====
// Alternative Initialization at Declaration
// =====
// Person person1 = {"Muhammad Adil", 20, "UET Peshawar"};

// =====
// Display Structure Members using Dot Operator
// =====
cout << "===== PERSON INFORMATION =====" << endl;
cout << "Name:    " << person1.name << endl;
cout << "Age:      " << person1.age << " years" << endl;
cout << "Address:  " << person1.address << endl;
cout << "===== " << endl;

return 0;
}

```

### Expected Output:

```

===== PERSON INFORMATION =====
Name:    Muhammad Adil
Age:      20 years
Address:  UET Peshawar, Pakistan
=====

```

## 3.2 Structure Declaration and Initialization (Student)

Declare a structure called Student with the following members: name (string), age (integer), grade (char). Declare a variable of type Student called student1 and initialize its members with values of your choice. Display values of student1 structure.

### Source Code:

```

#include <iostream>
#include <string>
using namespace std;

// =====
// Structure Declaration: Student

```

```

// =====
// Student structure to store student information
struct Student
{
    string name;    // Student's full name
    int age;        // Student's age in years
    char grade;     // Student's grade (A, B, C, D, F)
};

int main()
{
    // =====
    // Structure Variable Declaration
    // =====
    Student student1;

    // =====
    // Initialize Structure Members using Dot Operator
    // =====
    student1.name = "Muhammad Adil";    // Assign student name
    student1.age = 20;                  // Assign student age
    student1.grade = 'A';                // Assign grade (A = Excellent)

    // =====
    // Alternative: Initialize at declaration
    // Student student1 = {"Muhammad Adil", 20, 'A'};

    // =====
    // Display Student Information
    // =====
    cout << "===== STUDENT INFORMATION =====" << endl;
    cout << "Student Name: " << student1.name << endl;
    cout << "Age:          " << student1.age << " years" << endl;
    cout << "Grade:         " << student1.grade << endl;

    // =====
    // Display Grade Interpretation
    // =====
    cout << "\nGrade Interpretation: ";
    switch(student1.grade)
    {
        case 'A': cout << "Excellent"; break;
        case 'B': cout << "Good"; break;
        case 'C': cout << "Average"; break;
        case 'D': cout << "Below Average"; break;
        case 'F': cout << "Fail"; break;
        default: cout << "Invalid Grade";
    }
    cout << endl;
}

```

```

        cout << "===== " << endl;

        return 0;
    }

```

### Expected Output:

```

===== STUDENT INFORMATION =====
Student Name: Muhammad Adil
Age:          20 years
Grade:        A

Grade Interpretation: Excellent
=====

```

### 3.3 Accessing Structure Elements (Book)

Declare a structure called Book with the following members: title (string), author (string), price (float), pages (integer). Declare variables of type Book called book1 and book2, and initialize their members with values of your choice. Display values of structure members of variables book1 and book2. Also demonstrate structure copying.

#### Source Code:

```

#include <iostream>
#include <string>
using namespace std;

// =====
// Structure Declaration: Book
// =====
// Book structure to store book information
struct Book
{
    string title;      // Book title
    string author;     // Author name
    float price;       // Book price in dollars
    int pages;         // Number of pages
};

int main()
{
    // =====
    // Declare Two Structure Variables
    // =====
    Book book1; // First book variable
    Book book2; // Second book variable

    // =====

```

```

// Initialize First Book using Dot Operator
// =====
book1.title = "C++ Programming";
book1.author = "Bjarne Stroustrup";
book1.price = 45.99;
book1.pages = 1368;

// =====
// Initialize Second Book using Dot Operator
// =====
book2.title = "Data Structures";
book2.author = "Mark Allen Weiss";
book2.price = 39.50;
book2.pages = 672;

// =====
// Display First Book Information
// =====
cout << "===== BOOK 1 INFORMATION =====" << endl;
cout << "Title:  " << book1.title << endl;
cout << "Author: " << book1.author << endl;
cout << "Price:  $" << book1.price << endl;
cout << "Pages:  " << book1.pages << endl;

// =====
// Display Second Book Information
// =====
cout << "\n===== BOOK 2 INFORMATION =====" << endl;
cout << "Title:  " << book2.title << endl;
cout << "Author: " << book2.author << endl;
cout << "Price:  $" << book2.price << endl;
cout << "Pages:  " << book2.pages << endl;

// =====
// Demonstrate Structure Assignment (Copying)
// =====
// One structure variable can be assigned to another
// This copies ALL members from one variable to another
cout << "\n===== STRUCTURE COPYING =====" << endl;
cout << "Copying book1 to book3..." << endl;

Book book3 = book1;  // Copy all members from book1 to book3

cout << "\nBook 3 (Copy of Book 1):" << endl;
cout << "Title:  " << book3.title << endl;
cout << "Author: " << book3.author << endl;
cout << "Price:  $" << book3.price << endl;
cout << "Pages:  " << book3.pages << endl;
cout << "===== " << endl;

```

```

        return 0;
    }

```

### Expected Output:

```

===== BOOK 1 INFORMATION =====
Title:  C++ Programming
Author: Bjarne Stroustrup
Price:  $45.99
Pages:  1368

===== BOOK 2 INFORMATION =====
Title:  Data Structures
Author: Mark Allen Weiss
Price:  $39.5
Pages:  672

===== STRUCTURE COPYING =====
Copying book1 to book3...

Book 3 (Copy of Book 1):
Title:  C++ Programming
Author: Bjarne Stroustrup
Price:  $45.99
Pages:  1368
=====

```

## 3.4 Demonstration: Employee Information using Structure

Create a Person structure and take employee information as input from user using cin.get() for name (to handle spaces) and cin for numeric values. Display the entered information.

### Source Code:

```

#include <iostream>
#include <string>
using namespace std;

// =====
// Structure Declaration: Person
// =====
// Person structure for employee information
struct Person
{
    char name[50];    // Employee name (C-string for cin.get)
    int age;          // Employee age
    float salary;     // Employee salary
};

```

```

int main()
{
    // =====
    // Structure Variable Declaration
    // =====
    Person p1; // Variable to store employee information

    // =====
    // Input Section
    // =====
    cout << "===== EMPLOYEE INFORMATION SYSTEM =====" << endl;

    // Get employee name using cin.get to read full name with spaces
    cout << "Enter Full Name: ";
    cin.get(p1.name, 50); // Read up to 49 characters
    cin.ignore();         // Clear input buffer (remove newline)

    // Get employee age
    cout << "Enter Age: ";
    cin >> p1.age;

    // Get employee salary
    cout << "Enter Salary: $";
    cin >> p1.salary;

    // =====
    // Display Information
    // =====
    cout << "\n===== DISPLAYING INFORMATION =====" << endl;
    cout << "Name:    " << p1.name << endl;
    cout << "Age:     " << p1.age << " years" << endl;
    cout << "Salary: $" << p1.salary << endl;
    cout << "===== " << endl;

    return 0;
}

```

### Expected Output:

```

===== EMPLOYEE INFORMATION SYSTEM =====
Enter Full Name: Muhammad Adil
Enter Age: 20
Enter Salary: $5000

===== DISPLAYING INFORMATION =====
Name:    Muhammad Adil
Age:     20 years

```



Salary: \$5000

=====

## 4. CONCLUSION

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This lab demonstrated structures in C++, a powerful feature for grouping related data of different types. We learned how to declare structures using the `struct` keyword, define structure variables, initialize members using the dot operator, and access them for display. Structures enable better organization of complex data and are fundamental for object-oriented programming concepts. The ability to copy entire structures with a single assignment (`book3 = book1`) makes data management efficient. Structures can be nested, passed to functions, and form the basis for more advanced concepts like classes in C++.